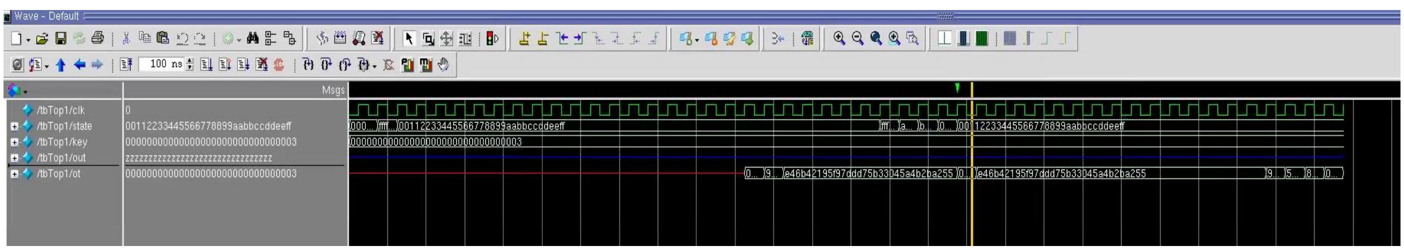
TROJAN 1: FUNCTIONAL TROJAN

Trojan 1 is a functional trojan. The objective of this trojan is to change the AES functionality of the circuit. The benchmark we used was AES-T1000 to implement the trojan. The trojan is activated when the following continuous sequence is inputted:

128'hFFFF_FFFF_FFFF_FFFF_FFFF_FFFF_FFFF_FFFF
128'hAAAA_AAAA_AAAA_AAAA_AAAA_AAAA_AAAA_AAAA
128'hBBBB_BBBB_BBBB_BBBB_BBBB_BBBB_BBBB_BBBB
128'h0000_0000_0000_0000_0000_0000_0000_0000

The trojan was implemented in multiple files attached to this report under *Trojan_1*. The file *trojan.v* is where the trigger and payload are located. In *trojan.v*, after the continuous sequence is detected, the current key of the AES algorithm is then copied and stored into an output register. Next, in the *top1.v* file the AES and trojan are connected to a selector. The selector chooses the AES output or the trojan's stored key depending on the trigger value of the trojan. Once triggered, the stored key is leaked to the output. This can be seen in the waveform below where the key bits are output after the continuous sequence is detected. As soon as the input state changes, the trigger is reset and the expected AES output continues.

Waveform:

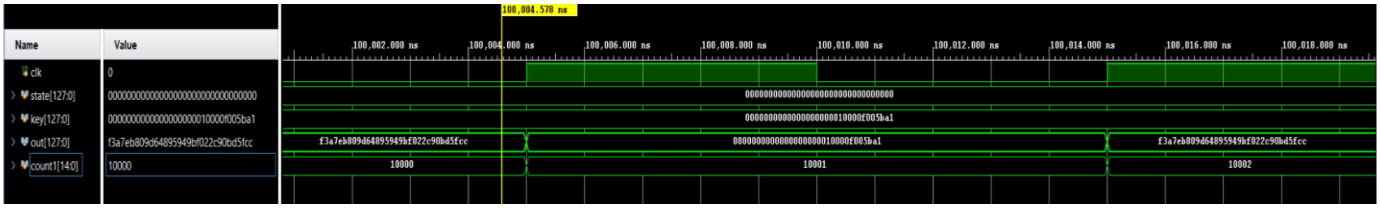


TROJAN 2: FUNCTIONAL TROJAN (TIMING DEPENDENT)

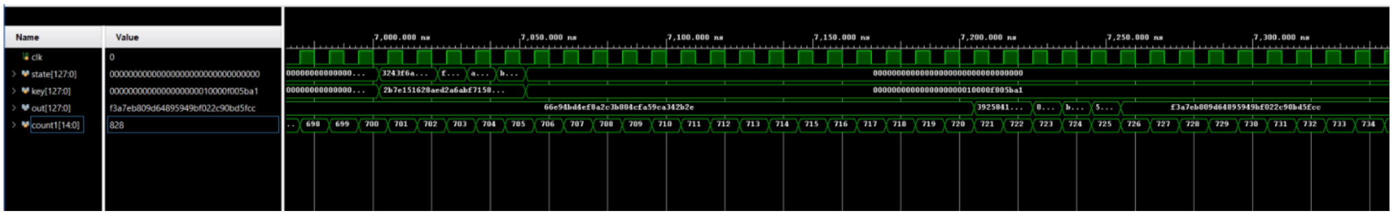
Trojan 2 is a functional trojan which is timing dependent. The trojan is triggered if it meets the two conditions: A counter counts to 10,000 and the trigger condition from trojan 1 is achieved. We implemented a synchronous counter which runs at the same clock as the AES hardware. As the counter reaches 10,000, the trigger condition of trojan 1 is checked. The payload of trojan 2 is the same as trojan 1.

Waveforms 1 & 2 below show the proper output of Trojan 2. In waveform 2 the continuous sequence from Trojan 1 is received. At this point the counter is at value 706/10000 and the payload is not released. Then in waveform 1 the counter reaches 10000 and in the following cycle on count 10001 the key bits are released to the output in place of the expected AES output.

Waveform 1:



Waveform 2:

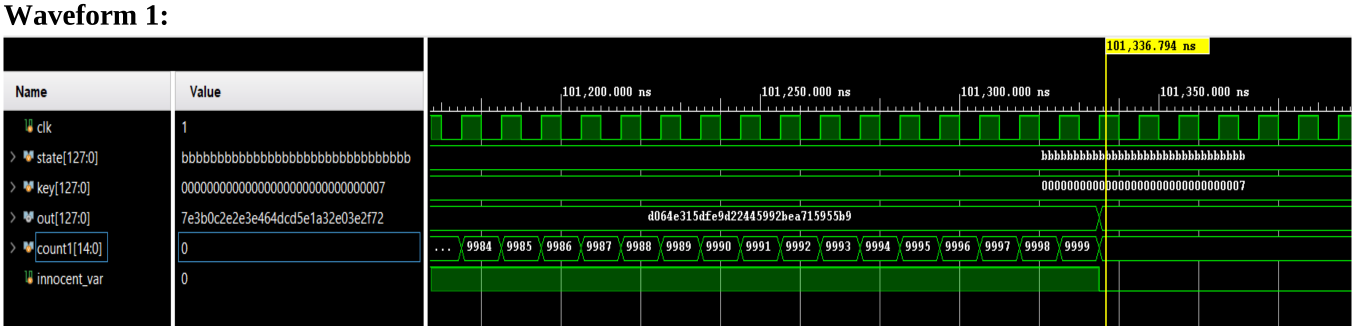


TROJAN 3: DEGRADATION-ATTACK TROJAN

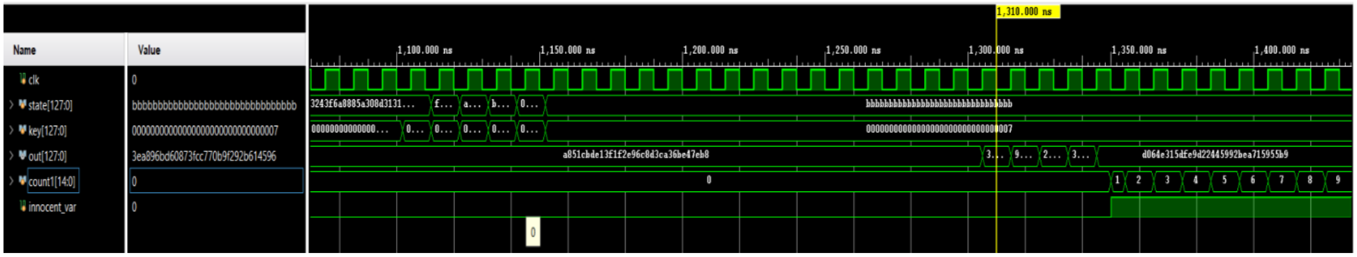
Trojan 3 is a degradation-attack trojan which ultimately degrades the performance of the circuit. When the trigger condition of trojan 1 is reached, the AES hardware on the performance level will slow down. For the payload, the trojan counts to 10,000 as seen in trojan 2. Note, for trojan 3 the output of the key is not revealed. The purpose of the attack is mainly to slow down the circuit.

In the aes_128.v file there is the trigger condition from trojan1. Furthermore, there is the counter (count1) that counts to 10000 once the trigger is triggered during this time the trojan delays the output for about 100,000 ns (with a 10ns clock cycle) the condition for this is in the final round method. After this period, it updates as expected since the payload is delivered.

Waveform 1:



Waveform 2:



Waveform 3:

(Benign)